

Shreya Sinha

+91 9308072415 | shreyasinha22052003@gmail.com

[linkedin.com/in/shreya-sinha-927626225](https://www.linkedin.com/in/shreya-sinha-927626225) | github.com/Shreya22-sinha | shreya-sinha.vercel.app

Summary

Backend engineer moving into product, with a habit of framing technical decisions as user and business trade-offs. Brings data fluency and close rapport with engineering teams to discovery, prioritisation and outcome-driven delivery.

Education

Birla Institute of Technology, Mesra	CGPA: 8.23
<i>Master of Computer Applications (MCA)</i>	2024 – 2026
Patna Women's College	CGPA: 9.01
<i>Bachelor of Computer Applications (BCA) — Platinum Jubilee Scholarship, all six semesters</i>	2021 – 2024

Technical and Product-Analytical Experience

Network People Services Technologies (NPST)	India
<i>Associate Software Developer</i>	Jul 2026 – Present
- Designed authentication and role-based access control for a 54-feature banking platform with mobile, web, and backend teams; delivered OTP/biometric login, 3 admin roles, and early maker-checker approval, enabling compliance traceability and reusable payment controls. - Built a bill-splitting product in a 2-day assessment to reduce shared-expense disputes; prioritized automated per-member balances and clear error handling, meeting all 6 core requirements. - Completed a 6-department cross-functional rotation and led problem framing for a 22-member team, aligning differing viewpoints into one recommendation.	

Research Publications

- Journal:** S. Sinha, L. Hota, K. K. Senapati, and A. Kumar, "SmartTraffic AI: A Real-Time Adaptive Traffic Signal Control System," *IEEE Transactions on Networking*, 2026. [Communicated]
Tackled urban congestion with a camera-only adaptive signal system, cutting simulated intersection wait time by 38.1% on low-cost hardware.
- Conference:** S. Sinha, L. Hota, and K. K. Senapati, "SmartTraffic AI for a Real-Time Adaptive Traffic Signal Control System," *2026 IEEE Future Networks World Forum (FNWF)*, 2026. [Communicated]
Improved operator trust in AI signal decisions via plain-language local explanations, reaching 87% confidence in 200 feedback rounds.
- Conference:** S. Sinha, L. Hota, K. K. Senapati, and P. Paul, "A Real-Time AI-Driven Traffic Management System using Deep Vision and Context-Aware AI," *4th International Conference on Innovation in Engineering & Management (ICIEM'26)*, BIT Noida, 2026. [Accepted]
Turned traffic images into congestion alerts and rerouting advice, choosing YOLOv8 after benchmarking 4 models (96.1% mAP@50).

Projects

- Vibeshelf** | *Express.js, React, Llama 3.3-70B (Groq), Jaccard Similarity* | [Devpost](#) Jun 2026
- At Mind the Product's **World Product Day hackathon** (2-person team), **led the Express.js backend** for cross-media vibe search, building auth, shelf/list APIs and a feedback-weighted re-ranker tuned against overcorrection, shipping a **live engine matching plain-language moods to 547 books, films and games**.
- Fraud Detection System** | *Python, FastAPI, Data Pipelines, Anomaly Detection* | [GitHub](#) Jan 2026 – Feb 2026
- With fraud at **2% of ~100K transactions** (a flag-nothing model scores 98%), **benchmarked Isolation Forest, One-Class SVM and an autoencoder** on precision, recall, F1 and AUC-ROC, set the threshold by weighing missed-fraud losses against review load, and **shipped it as a production-ready REST API** with validation and logging.
- Smart AI Traffic Management System** | *YOLOv8, PyTorch, Flask, LLaMA 3* | [GitHub](#) Feb 2026 – Mar 2026
- After a **Deep Q-Network** (~90% detection accuracy) **cut wait time 30% and raised throughput 25%** vs. a fixed timer, **prioritised operator adoption over further tuning** with a local LLaMA 3 layer explaining every signal decision in plain language, leading to an **ICIEM'26 acceptance**.

Leadership & Volunteering

School Mentorship Program, Patna Women's College	Patna, India
<i>Volunteer Teacher</i>	
- Volunteered to close learning gaps at a government all-girls school, owning instruction for 120+ students across Classes 7–11 with widely varying learning levels. - Adapted lesson plans to each grade's level and pace and iterated on delivery using classroom feedback, improving the student pass rate by 10%.	

Skills

Product & Analysis: Metric Definition, Trade-off Analysis, Success Criteria, Documentation, Stakeholder Communication, SQL
Languages & Data: Python, SQL, JavaScript, Java, C++; PostgreSQL, MySQL, MongoDB
Backend & Systems: FastAPI, Flask, REST APIs, Spring Boot, React, Linux, Git, Docker, Postman
Data & ML: Data Pipelines, PyTorch, Scikit-learn, Anomaly Detection, Computer Vision, Reinforcement Learning, NLP